LECTURE 7:
HIERARCHICAL BAYES FOR ANOVA

Key concepts:

- F-test for pooling
- Posterior distribution for HB ANOVA
- Improper posterior
- Uniform prior on standard deviation
- Half-Cauchy prior on standard deviation

1. Introduction

In a one-way ANOVA model, observations are normally distributed with different means in each group. An F-test can be used to test whether the means differ and, if not, the data can be pooled. The Hierarchical Bayesian approach is to treat the means as random variables coming from a common distribution. The pooled (equal means) model implies that the distribution of means has zero variance which seems unlikely. However, the HB approach generally improves upon separate estimators (a phenomenon known as the "Stein effect," one of the more remarkable results of modern statistics).

We provide a heuristic introduction to Bayesian ANOVA first assuming that the unit level variances are known.

2. Classical ANOVA

A one-way Analysis of Variance (ANOVA) for y_{ji} ($j = 1, \ldots, J$; $i = 1, \ldots, n_j$) uses the model

$$y_{j1}, \ldots, y_{jn_j} \sim \text{Normal}(\theta_j, \sigma^2)$$

for $j = 1, \ldots, J$ groups. The MLE for θ_j is

$$\bar{y}_{j\cdot} = \frac{1}{n_j} \sum_{i=1}^{n_i} y_{ji},$$

where $\bar{y}_{j\cdot} \sim \text{Normal}(\theta_j, \sigma_j^2)$ and $\sigma_j^2 = \sigma^2/n_j$. The *grand mean* is

$$\bar{y}_{\cdot\cdot} = \frac{\sum_{j=1}^{J} \sum_{i=1}^{n_i} y_{ij}}{\sum_{j=1}^{J} n_j} = \frac{\sum_{j=1}^{J} n_j \bar{y}_{j\cdot}}{J\bar{n}}$$

where $\bar{n} = \sum_{j=1}^{J} n_j/J$. The design is said to be *balanced* if $n_j = n$ for $j = 1, \ldots, J$, in which case $\bar{y}_{\cdot\cdot} = \sum_{j=1}^{J} \bar{y}_{j\cdot}/J$. In practice, designs are rarely exactly balanced, but the concepts are easier to understand for balanced designs. The corresponding ANOVA table shown below assumes balance.

Source	d.f.	Sum of squares	Mean square	Expected mean square
Between groups	$J-1$	$\sum_{j=1}^{J} n(\bar{y}_{j\cdot} - \bar{y}_{\cdot\cdot})^2$	SS$/(J-1)$	$\sigma^2 + \sum_{j=1}^{J}(\theta_i - \bar{\theta})^2/J$
Within groups	$J(n-1)$	$\sum_{j=1}^{J}\sum_{i=1}^{n}(y_{ji} - \bar{y}_{j\cdot})^2$	SS$/[J(n-1)]$	σ^2
Total	$Jn-1$	$\sum_{j=1}^{J}\sum_{i=1}^{n_i}(y_{ji} - \bar{y}_{\cdot\cdot})^2$	SS$/(Jn-1)$	

To test $H_0 : \theta_1 = \cdots = \theta_J$, we form the F-statistic,

$$F = \frac{\text{Between MS}}{\text{Within MS}} = \frac{\sum_{j=1}^{J} n(\bar{y}_{j\cdot} - \bar{y}_{\cdot\cdot})^2/(J-1)}{\sum_{j=1}^{J}\sum_{i=1}^{n}(y_{ji} - \bar{y}_{j\cdot})^2/[J(n-1)]}$$

Under H_0, both the numerator and denominator are estimates of σ^2, while under the alternative, the numerator tends to be larger. Let $F_{J-1,J(n-1)}(\alpha)$ be the α quantile of the F distribution with $J-1$ and $J(n-1)$ degrees of freedom. We reject H_0 at significance level α if $F > F_{J-1,J(n-1)}(1-\alpha)$; otherwise, we do not reject. Normally, if we do not reject, we then *pool* the data from the J groups and estimate each θ_j by $\bar{y}_{\cdot\cdot}$. If we reject, then we use the separate (unpooled) estimators $\bar{y}_{j\cdot}$ for each θ_j.

3. Bayesian ANOVA

The Bayesian model has three components:

Data model $\bar{y}_{1\cdot}, \ldots, \bar{y}_{J\cdot} | \theta_1, \ldots, \theta_J \overset{\text{ind}}{\sim} \text{Normal}(\theta_j, \sigma_j^2)$

Prior $\theta_1, \ldots, \theta_J \sim$ i.i.d. Normal(μ, τ^2)

Hyperprior $(\mu, \tau^2) \sim p(\mu, \tau^2)$

A few notes:

- The data model is conditional upon all of the parameters, including the hyperparameters. The fact that the hyperparameters (μ, τ^2) do not appear in the data model implies that the data are independent of the hyperparameters conditional upon the lower level parameters.
- While not stated explicitly, the data from different groups are independent of one another conditional upon the parameters.
- We will use either an improper uniform prior for μ ($p(\mu) \propto 1$) or an informative normal prior ($\mu \sim \text{Normal}(\mu_0, \omega_0^2)$). Most of the hard work involves choosing a prior for the second level variance τ^2, which we will discuss in detail.
- In many applications, as here, the data variances σ^2 are assumed equal, though the number of observations per group n_j varies. Initially, we will just assume that the data variances are known.
- In the schools example in Chapter 5 of *BDA3*, there is a single observation per group ($n_j = 1$ for $j = 1, \ldots, J$). If you read the notes carefully, this is a regression coefficient and σ_j is its (estimated) standard error.

3.1. **Prior Distributions.** We will use the following prior pdfs:

$$p(\theta_1, \ldots, \theta_J | \mu, \tau^2) = \prod_{j=1}^{J} p(\theta_i | \mu, \tau^2) \propto \tau^{-n} \exp\left\{ -\frac{\sum_{j=1}^{J} (\theta_j - \mu)^2}{2\tau^2} \right\}$$

$$p(\mu, \tau) = p(\mu|\tau)\, p(\tau) \propto p(\tau)$$

3.2. **Likelihood Function.** The likelihood is normal with known variance.

$$p(y|\theta) \propto \prod_{j=1}^{J} \sigma_j^{-1} \exp\left\{ -\frac{(\bar{y}_{j\cdot} - \theta_j)^2}{2\sigma_j^2} \right\}$$

This will generalize later when there are n_j observations in group j, each with variance σ^2. Then $\bar{y}_{j\cdot}$ will be a sufficient statistic with variance $\sigma_j^2 = \sigma^2/n_j$. It is also feasible to allow the variances to differ across groups.

3.3. **Marginal Posterior Distribution of Hyperparameters.** This problem is unusual in that the distribution of the data given the hyperparameters is available:

$$\bar{y}_{1\cdot}, \ldots, \bar{y}_{J\cdot} | \mu, \tau^2 \overset{\text{ind}}{\sim} \text{Normal}(\mu, \tau^2 + \sigma_j^2),$$

so

$$p(y|\mu, \tau^2) \propto \prod_{j=1}^{J} (\tau^2 + \sigma_j^2)^{-1/2} \exp\left\{ -\frac{(\bar{y}_{j\cdot} - \mu)^2}{2(\tau^2 + \sigma_j^2)} \right\}.$$

This result is used in the next section.

3.4. **Joint Posterior of All Parameters.** Gelman *et al.* (2013), p. 116 skip the algebra and reason as follows: treating (μ, τ^2) as given, we have a normal likelihood for $\bar{y}_{j\cdot}$ with mean θ_j and known variance σ_j^2; the prior distribution for θ_j is Normal(μ, τ^2), so we can use the result from Lecture 3 to get the (conditional) posterior: $\theta_j | \mu, \tau^2, y \sim \text{Normal}(\hat{\theta}_j, V_j)$.

Here is an algebraic version of the same result starting from the beginning. The joint posterior is proportional to

$$\begin{aligned}
p(\theta, \mu, \tau | y) &\propto p(\mu, \tau)\, p(\theta|\mu, \tau)\, p(y|\theta) \\
&\propto p(\mu, \tau) \prod_{j=1}^{J} \tau^{-1} \exp\left\{ -\frac{(\theta_j - \mu)^2}{2\tau^2} \right\} \sigma_j^{-1} \exp\left\{ -\frac{(\bar{y}_{j\cdot} - \theta_j)^2}{2\sigma_j^2} \right\} \\
&\propto p(\tau)\, \tau^{-J} \exp\left\{ -\frac{1}{2} \sum_{j=1}^{J} \left[\frac{(\theta_j - \mu)^2}{\tau^2} + \frac{(\bar{y}_{j\cdot} - \theta_j)^2}{\sigma_j^2} \right] \right\} \\
&\propto p(\tau)\, \tau^{-J} \exp\left\{ -\frac{1}{2} \sum_{j=1}^{J} \left[\left(\frac{1}{\tau^2} + \frac{1}{\sigma_j^2} \right) \theta_j^2 - 2 \left(\frac{\mu}{\tau^2} + \frac{\bar{y}_{j\cdot}}{\sigma_j^2} \right) \theta_j \right] \right\} \exp\left\{ -\frac{J\mu^2}{2\tau^2} \right\} \\
&\propto p(\tau)\, \tau^{-J} \exp\left\{ -\frac{1}{2} \sum_{j=1}^{J} \left(\frac{\mu^2}{\tau^2} - \frac{\hat{\theta}_j^2}{V_j} \right) \right\} \prod_{j=1}^{J} \exp\left\{ -\frac{(\theta_j - \hat{\theta}_j)^2}{2V_j} \right\}
\end{aligned}$$

where

$$\hat{\theta}_j = \frac{\dfrac{\mu}{\tau^2} + \dfrac{\bar{y}_{j\cdot}}{\sigma_j^2}}{\dfrac{1}{\tau^2} + \dfrac{1}{\sigma_j^2}} \qquad V_j = \left(\frac{1}{\tau^2} + \frac{1}{\sigma_j^2} \right)^{-1}$$

Some tedious algebra shows that

$$\frac{\mu^2}{\tau^2} - \frac{\hat{\theta}_j^2}{V_j} = \frac{\mu^2 - 2\mu\bar{y}_{j\cdot} - \tau^2\bar{y}_{j\cdot}^2/\sigma_j^2}{\tau^2 + \sigma_j^2}$$

so

$$\begin{aligned}
\sum_{j=1}^{J} \left(\frac{\mu^2}{\tau^2} - \frac{\hat{\theta}_j^2}{V_j} \right) &= \mu^2 \sum_{j=1}^{J} \frac{1}{\tau^2 + \sigma_j^2} - 2\mu \sum_{j=1}^{J} \frac{\bar{y}_{j\cdot}}{\tau^2 + \sigma_j^2} - \tau^2 \sum_{j=1}^{J} \frac{\bar{y}_{j\cdot}^2}{\sigma_j^2(\tau^2 + \sigma_j^2)} \\
&= V_\mu^{-1}(\mu - \hat{\mu})^2 - V_\mu^{-1}\hat{\mu}^2 - \tau^2 \sum_{j=1}^{J} \frac{\bar{y}_{j\cdot}^2}{\sigma_j^2(\tau^2 + \sigma_j^2)},
\end{aligned}$$

where

$$\hat{\mu} = \frac{\displaystyle\sum_{j=1}^{J} \frac{\bar{y}_{j\cdot}}{\tau^2 + \sigma_j^2}}{\displaystyle\sum_{j=1}^{J} \frac{1}{\tau^2 + \sigma_j^2}} \qquad V_\mu = \left(\sum_{j=1}^{J} \frac{1}{\tau^2 + \sigma_j^2} \right)^{-1}$$

This shows that the posterior factors into

$$\begin{aligned}
p(\theta, \mu, \tau | y) &\propto q(\tau, y) V_\mu^{-1/2} \exp\left\{ -\frac{(\mu - \hat{\mu})^2}{2V_\mu} \right\} V_j^{-1/2} \exp\left\{ -\frac{(\theta_j - \hat{\theta}_j)^2}{2V_j} \right\} \\
&\propto q(\tau, y)\, p(\mu|\tau, y) \prod_{j=1}^{J} p(\theta_j|\mu, \tau, y)
\end{aligned}$$

where

$$\begin{aligned}
p(\mu|\tau, y) &\propto V_\mu^{-1/2} \exp\left\{ -\frac{(\mu - \hat{\mu})^2}{2V_\mu} \right\} \\
p(\theta_j|\mu, \tau, y) &\propto V_j^{-1/2} \exp\left\{ -\frac{(\theta_j - \hat{\theta}_j)^2}{2V_j} \right\}
\end{aligned}$$

or

$$\mu|\tau, y \sim \text{Normal}(\hat{\mu}, V_\mu) \qquad \theta_1, \ldots, \theta_J|\mu, \tau, y \overset{\text{ind}}{\sim} \text{Normal}(\hat{\theta}_j, V_j)$$

Finally, to find the marginal posterior for τ, we use

$$p(\tau|y) = \frac{p(\mu,\tau|y)}{p(\mu|\tau,y)}$$
$$\propto \frac{p(\tau)\prod_{j=1}^{J}(\sigma_j^2+\tau^2)^{-1/2}\exp\left\{-(\bar{y}_{j\cdot}-\mu)^2/2(\sigma_j^2+\tau^2)\right\}}{V_\mu^{-1/2}\exp\left\{-(\mu-\hat{\mu})^2/2V_\mu\right\}}.$$

This holds for all μ, so we may substitute $\mu=\hat{\mu}$ to obtain

$$p(\tau|y) \propto p(\tau)\,V_\mu^{1/2}\prod_{j=1}^{J}\left(\sigma_j^2+\tau^2\right)^{-1/2}\exp\left\{-\frac{(\bar{y}_{j\cdot}-\hat{\mu})^2}{2(\sigma_j^2+\tau^2)}\right\}.$$

Next, we turn to priors for τ.

4. Priors for Variance Parameters

Previously, we showed that

$$p(\tau|y) \propto p(\tau)\,V_\mu^{1/2}\prod_{j=1}^{J}(\sigma_j^2+\tau^2)^{-1/2}\exp\left\{-\frac{(\bar{y}_{\cdot j}-\hat{\mu})^2}{2(\sigma_j^2+\tau^2)}\right\}$$

A number of different priors have been suggested for τ.

4.1. **Uniform Prior on Log Scale.** The obvious choice for a non-informative prior is

$$p(\log\tau) \propto 1$$

which is equivalent to

$$p(\tau) \propto \frac{1}{\tau}.$$

Unfortunately, this leads to an improper posterior. The unnormalized posterior is

$$q(\tau) = \frac{1}{\tau}\,V_\mu^{1/2}\prod_{j=1}^{J}(\sigma_j^2+\tau^2)^{-1/2}\exp\left\{-\frac{(\bar{y}_{\cdot j}-\hat{\mu})^2}{2(\sigma_j^2+\tau^2)}\right\}$$

For $0<\tau\leq 1$, we have $\sigma_j^2+\tau^2\leq K+1$ (where $K=\max_{1\leq j\leq J}\sigma_j^2$), so

$$(\sigma_j+\tau^2)^{1/2} \geq (K+1)^{-1/2}.$$

Also, for $\tau>0$, we have $\sigma_j^2+\tau^2\geq k$ (where $k=\min_{1\leq j\leq J}\sigma_j^2$), so

$$\exp\left\{-\frac{(\bar{y}_{j\cdot}-\hat{\mu})^2}{2(\sigma_j^2+\tau^2)}\right\} \geq \exp\left\{-\frac{(\bar{y}_{j\cdot}-\hat{\mu})^2}{2k}\right\}.$$

Combining these results shows that

$$\int_0^\infty q(\tau)\,d\tau \geq \int_0^1 q(\tau)\,d\tau \geq \frac{1}{\tau} V_\mu^{1/2} \prod_{j=1}^{J} (K+1)^{-1/2} \exp\left\{-\frac{(\bar{y}_{j\cdot} - \hat{\mu})^2}{2k}\right\} d\tau$$
$$= V_\mu^{1/2} (K+1)^{-J/2} \exp\left\{-\frac{\sum_{j=1}^{J} (\bar{y}_{j\cdot} - \hat{\mu})^2}{2k}\right\} \int_0^1 \frac{d\tau}{\tau} = \infty.$$

This is a typical method for showing that an unnormalized distribution is improper.

4.2. **Uniform Prior on the Standard Deviation.** Instead, we try an improper uniform prior on the standard deviation,

$$p(\tau) \propto 1 \quad \text{for } \tau > 0$$

A similar argument now shows that the posterior is proper. Denote the unnormalized posterior by

$$q(\tau) = V_\mu^{1/2} \prod_{j=1}^{J} (\sigma_j^2 + \tau^2)^{-1/2} \exp\left\{-\frac{(\bar{y}_{\cdot j} - \hat{\mu})^2}{2(\sigma_j^2 + \tau^2)}\right\}.$$

This time, we bound the terms below: $(\sigma_j^2 + \tau^2)^{-1/2} \leq (k + \tau^2)^{-1/2}$ and

$$\exp\left\{-\frac{(\bar{y}_{j\cdot} - \hat{\mu})^2}{2(\sigma_j^2 + \tau^2)}\right\} \leq \exp\left\{-\frac{(\bar{y}_{j\cdot} - \hat{\mu})^2}{2(K + \tau^2)}\right\}.$$

so that

$$\int_0^\infty q(\tau)\,d\tau \leq V_\mu^{1/2} \int_0^\infty \prod_{j=1}^{J} (k + \tau^2)^{-1/2} \exp\left\{-\frac{(\bar{y}_{j\cdot} - \hat{\mu})^2}{2(K + \tau^2)}\right\} d\tau$$
$$= V_\mu^{1/2} \int_0^\infty (k + \tau^2)^{-J/2} \exp\left\{-\frac{\sum_{j=1}^{J} (\bar{y}_{j\cdot} - \hat{\mu})^2}{2(K + \tau^2)}\right\} d\tau < \infty$$

(Why?) This proves that the posterior is proper with a uniform prior on the standard deviation τ.

4.3. **Half Cauchy Prior for Standard Deviation.** Gelman (2007) suggested using a Half Cauchy prior on the standard deviation. The Cauchy distribution is a t distribution with one degree of freedom. The Half Cauchy is the Cauchy distribution truncated to be positive, *i.e.*

$$p(y) \propto \left[1 + \left(\frac{y - \mu}{\sigma}\right)^2\right]^{-1/2} \mathbb{1}(y > 0)$$

In JAGS, you can get this distribution using `y ~ dt(mu, sigma, 1)T(0,)`.

In Stan, use `real<lower=0> y;` and `y ~ cauchy(mu, sigma);` to draw a Half Cauchy.